

Standard Operating Procedure for Measuring Electrical Conductivity

1. First step is to fill in the electrical conductivity (EC) instrument calibration log form.
2. The standard to use as a check is $1000 \pm 10 \mu\text{S/cm}$ standard solution.
3. Pour the standard solution into a clean (dry and dust free) plastic bottle and label it.
4. Turn the meter on.
5. Be sure the meter is in the conductivity measurement mode.
6. Rinse the probe well with distilled water and gently dry with soft tissue.
7. Place the probe into the standard solution in the plastic bottle.
8. Make sure the probe is immersed well into the solution so that the slot at the bottom of the probe is totally immersed in the solution.
9. Agitate the sample with the probe for a few seconds to remove bubbles that may be trapped in the slot.
10. Press the "Read" key and wait for the reading to stabilise. The word stabilising will disappear from the screen when the reading is stable.
11. Check the reading of the standard solution. It should fall within $1000 \pm 10 \mu\text{S/cm}$ (i.e. 990 to 1010).
12. If the reading is outside this range then calibrate the meter using the procedure outlined below.
13. After re-calibration check with the standard solution again and record the measured value.
14. If the reading is within the acceptable range, then proceed with the sample measurements.

Procedure for calibrating the session5 conductivity meter

1. Place the sample probe into the $1000 \pm 10 \mu\text{S/cm}$ standard solution which was poured into a clean (dry and dust free) plastic bottle for the above steps.
2. Turn the meter on.
3. Be sure the meter is in the conductivity measurement mode.
4. Make sure the probe is immersed well into the solution so that the slot at the bottom of the probe is totally immersed in the solution.
5. Agitate the sample with the probe for a few seconds to remove bubbles that may be trapped in the slot.
6. Press CAL key.
7. The last calibration value will appear which is $1000 \mu\text{S/cm}$.
8. To calibrate, press ENTER.
9. When the reading is stable, the calibration will be automatically stored and the instrument returns to the reading mode.
10. The meter is ready for use and the above check as per the EC instrument calibration log form should be done.

Procedure for Measuring EC of Water Samples

1. Take out the EC result recording sheet and fill in sample details.
2. The first field to be filled asks whether the EC meter has been checked for calibration. This was done following the above 2 sets of procedures.
3. After the EC meter has been checked for calibration rinse the probe well with distilled water.
4. Place the probe in the water sample to be measured.
5. Agitate the sample with the probe for a few seconds to remove bubbles that may be trapped in the slot.
6. Press the "Read" key and wait for the reading to stabilise. The word stabilising will disappear from the screen when the reading is stable.
7. Record the readings in the appropriate column and note the UNITS from the instrument.

Note: To perform TDS or salinity measurements press the appropriate key (TDS or SAL) in the keypad and then perform the measurements. The calibration done using the conductivity mode also holds for the TDS and SAL mode.

Additional Information

For more detailed information on calibration and troubleshooting, refer to the “sension5 Conductivity meter” Manual provided.

Notes:

Electrical Conductivity

Electrical Conductivity (EC) is a measure of how much total salt (inorganic ions such as sodium, chloride, magnesium and calcium) is present in the water. The more ions, the higher the conductivity. Monitoring of this parameter is important in drinking water, especially for water supplies that are taken from **boreholes or wells on atoll islands** containing a freshwater lens on top of underlying salt water. If the water supply demand exceeds the capacity of the lens to replenish itself through rainfall, infiltration and recharge, the freshwater lens becomes thinner and increasing concentrations of salt may be observed. At high salt levels, consumers will detect an unpleasant taste, washing clothes will be difficult, the water may not quench thirst, and diarrhoea may occur.

Increasing conductivity over time in water indicates that one or more inorganic constituents are also increasing; this situation should trigger further investigations. For atoll islands using groundwater this could be for example, measuring the chloride content of water. Seawater intrusion in groundwater can be a cause of increased chlorides. Hence the pumping rates can be lowered or stopped altogether for the freshwater lens to replenish.

Guideline EC values to note:

Water bracket	EC value ($\mu\text{S}/\text{cm}$)
Freshwater	0 - 2,500
Brackish water	2,500 - 50,000
Seawater	> 50,000

Also note that EC values are NOT of health concern.